

Question Paper

Exam Date & Time: 17-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Principles of Computer Vision [CSE_3125]

Marks: 50

Duration: 180 mins.

PART-B

Answer all the questions.

- 1) Summarize the difference between the image processing and computer vision. Explain the components of visual perception in detail. (5)

A)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- B) Given a 1-D discrete signal and a filter, Apply both correlation and convolution operations. (5)

Let the input signal and filter be:

$$f = [0, 0, 1, 2, 3, 0, 0]$$

$$w = [1, 2, 3]$$

1. Perform correlation between f and w by aligning the filter to the input signal and sliding it across all positions (with zero-padding).
 - o Show all intermediate steps.
 - o Write the full correlation result and the cropped correlation result.
2. Perform convolution between f and w by rotating the filter 180° and then sliding it across the signal (with zero-padding).
 - o Show all intermediate steps.
 - o Write the full convolution result and the cropped convolution result.
3. Compare the two results and explain the key difference between correlation and convolution in this example.

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) Define thresholding. Using the Global Thresholding Algorithm, compute the threshold value T for the following 5×5 image (intensity values from 0 to 7), and show the thresholding result. (5)
- A)

1	4	6	3	5
0	2	7	4	1
3	5	6	2	4
1	3	5	6	0
2	4	7	3	2

Course Outcome: "CO2" / Cognitive Level: "Understand"

- B) Illustrate the objective of the Canny Edge Detection technique? Apply each stage of the Canny algorithm along with the mathematical equations used for edge detection. (5)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3) A mobile app that scans documents often captures images that are tilted or misaligned due to uneven camera angles. Explain how geometric transformations help in adjusting the scanned image's position, orientation, and shape to achieve proper alignment (5)
- A)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- B) An autonomous robot uses a camera to detect boxes in front of it. Due to lighting changes and background noise, some feature points are mismatched, causing detection errors. Apply the RANSAC algorithm to explain how it helps in building a reliable object model by eliminating outliers and improving feature matching accuracy. (5)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) A smart traffic monitoring system records video footage at a busy intersection to track vehicles and estimate their movement patterns. Apply the concept of translational alignment to explain how the system determines the direction and speed of moving vehicles across consecutive video frames. (5)
- A)

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) You must design a real-time video denoising system for a CCTV surveillance feed running on limited edge hardware (no GPU, low power, and restricted memory). The video suffers from noise, compression artifacts, and moderate object motion. To perform motion-compensated denoising, you must choose either dense optical flow or KLT-based feature tracking as the motion estimation backbone. Analyze which method is more suitable for this application and justify your decision in terms of accuracy, computational cost, robustness in low-texture regions or night-time scenes, Smoothness across frames, and impact on object detection. (5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) Apply machine learning-based approach to develop a tumor localization and classification technique for radiologists using MRI or CT images. How would you combine detection with medical image classification to reduce false negatives? Give detailed steps for the applied technique. (5)
- A)

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) A large e-commerce platform wants to enable "search by image", where a user can upload a photo of a product (e.g., shoes, bags, furniture) and instantly retrieve visually similar items from a catalogue of 10 million images. Apply techniques of computer vision to develop a framework to retrieve the most relevant images from the database. Provide description of the techniques in the applied framework. (5)

Course Outcome: "CO5" / Cognitive Level: "Application"

-----End-----

Event Name	Subject
------------	---------

058-END SEMSTER-ODD-2025 CSE_3125:Principles of Computer Vision

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
------------	---------

058-END SEMSTER-ODD-2025 CSE_3125:Principles of Computer Vision

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	3	15(30%)
3	Application	6	30(60%)
4	Analysis	1	5(10%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)